

## Education

### Tongji University, School of Chemical Science and Engineering

Sep. 2024 – Jun. 2028 (expected)

B.S. in Chemistry · National Top-Notch Student Training Program 2.0 Base for Basic Disciplines (Chemistry)

**GPA:** 4.53/5.00 (90.30/100) · **Rank:** 2/28 in the Top-Notch Program; top 5% in the School of Chemical Science and Engineering

**Selected coursework:** Inorganic Chemistry (Honors), Analytical Chemistry (Honors), Organic Chemistry, Physical Chemistry, Python Programming, and Foundations of Artificial Intelligence

## Publication

### Fluorenone Acceptor–Linker Engineering of Lead Halide MOFs for Aerobic Photocatalytic Oxidative Coupling

Dongyang Li, **Junkai Kang**, Fanyu Meng, Wenyan Dan\*, and Honghan Fei\*

*Inorganic Chemistry* (2026), accepted August 19, 2026 · DOI: [10.1021/acs.inorgchem.6c02797](https://doi.org/10.1021/acs.inorgchem.6c02797)

## Research Experience

### Molecular Design and Synthesis of Ortho-Substituted Tetraphenylethylene-Functionalized Non-Fullerene Acceptors toward Low-Energy-Loss Organic Solar Cells

Jul. 2026 – Present

Summer Research Student · Department of Chemistry, HKUST · Advisor: Prof. He Yan

- Investigating whether ortho substitution in tetraphenylethylene-derived terminal groups can restrict intramolecular motion and mitigate non-radiative voltage loss in Y-series non-fullerene acceptors without compromising packing or charge transport.
- Designing and synthesizing a parent/ortho-substituted candidate series through literature-guided multistep routes, with chromatographic and NMR checkpoints before photophysical and device evaluation.

### Halide-Dependent Structure and Optical Properties of Molecule-Connected Pb–Cu Chalcogenide Frameworks toward Photocatalytic CO<sub>2</sub> Reduction

Dec. 2025 – Present

Independent Undergraduate Project Lead · Tongji University · Advisor: Prof. Honghan Fei

- Developed a two-stage synthesis based on 2-mercaptopyridine that couples interfacial Cu–thiolate precursor formation with a solvothermal PbX<sub>2</sub> step; optimized acid, halide, and solvothermal conditions through PXRD-guided screening.
- Compared TJU-88(Br), [Pb<sub>2</sub>Cu<sub>2</sub>S<sub>2</sub>Br<sub>2</sub>], with TJU-88(I), [Pb<sub>2</sub>Cu<sub>2</sub>S<sub>2</sub>I<sub>2</sub>], using experimental PXRD, crystallographic models, and UV–vis absorption; replacing Br with I shifted the absorption edge from 546 to 557 nm and narrowed the optical band gap from 2.41 to 2.35 eV.

### ZIF-67 Nanozymes: Synthesis and Kinetic Analysis of Enzyme-Mimetic Activity

Sep. 2024 – Jan. 2025

Project Leader · Tongji University Innovative Experiment Design

- Prepared approximately 200 nm polyhedral ZIF-67 particles by room-temperature coordination self-assembly and characterized morphology and elemental distribution by SEM and elemental mapping.
- Quantified glucose oxidase-like and peroxidase-like activities by UV–vis kinetics and Michaelis–Menten/Lineweaver–Burk analysis; obtained an apparent Michaelis constant ( $K_{m,app}$ ) of 0.05 mM for TMB.
- Received First Prize (1st place) in the Shanghai Undergraduate Innovative Experiment Design Competition.

## Selected Honors and Awards

- **National Scholarship**, Ministry of Education of China (2025)
- **Third Prize**, National College Student Innovation Experience Competition (2024)
- **First Prize & Roadshow Champion**, Tongji University Innovation Experience Competition (2025)
- **Outstanding Student**, Tongji University (2025)
- **Third Prize**, “Guoyao Cup” Chemistry Competition (2024 and 2025)
- **Outstanding Camper**, Peking University Chemistry Camp (2022 and 2023; high school)
- **Gold Award**, China International College Students’ Innovation Competition, Tongji University (2026)
- **First Prize (1st place)**, Shanghai Undergraduate Innovative Experiment Design Competition (2025)
- **Tang Zhongying Scholarship**, Tongji University (2024 and 2025)
- **First Prize**, Tongji University Physics Competition (2025)
- **Third Prize**, China Undergraduate Mathematical Contest in Modeling (2024)
- **Third Prize**, Chinese Chemistry Olympiad (high school)

## Technical Skills

**Laboratory synthesis:** Schlenk-line and inert-atmosphere synthesis; solvothermal synthesis; multistep organic synthesis; column chromatography.

**Characterization and analysis:** XRD/PXRD and profile comparison; UV–vis spectroscopy; NMR spectroscopy; SEM/elemental-mapping interpretation; enzyme-kinetics analysis.

**Data and software:** Python (pandas, NumPy, Matplotlib, SciPy); OriginPro; ChemDraw; Mercury; LaTeX; scientific data analysis and visualization.

## Leadership and Service

- **Class President, Chemistry Top-Notch Program:** organized more than 30 class and university activities; two interdisciplinary education initiatives were selected as university key projects.
- **Peer Mentor, Qidi College:** supported junior students in academic planning, university transition, and early research exploration.
- **Student Union, Public Relations and External Liaison:** contributed to university-wide cultural events and intercollegiate coordination.
- **Service recognitions:** Outstanding Volunteer, Tongji University Student Union; Outstanding Class President and Outstanding League Cadre, School of Chemical Science and Engineering; Outstanding Individual, student Party branch initiative.
- **Community and science outreach:** completed approximately 200 hours across community education, environmental fieldwork, university outreach, and campus service.

## Additional Experience

**Technology Investment Intern, Bank of East Asia, Shanghai**

2025

### Academic Presentation and Science Communication

Delivered research presentations and chemistry-outreach sessions for high-school students in Henan (2024 and 2025).